

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

07
REGULAR EXPRESSION

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REVIEW

Finite Automata with Outputs:

Mealy Machine

Moore Machine

Conversion of Moore Machine to Mealy Machine

Conversion of Mealy Machine to Moore Machine

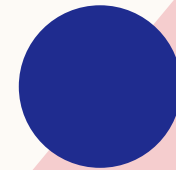
OUTLINE

Identities of Regular Expression

Arden's Theorem

Conversion of FSA to Regular Expression

Conversion of Regular Expression to FSA



REGULAR EXPRESSION

Regular expressions are used for representing certain sets of strings in an algebraic fashion.

1. Any terminal symbols (symbols $\in \Sigma$, including $\emptyset$) are regular expressions.
2. The union of two regular expressions is also a regular expression.
 - $R_1, R_2 \rightarrow R_1 + R_2$
3. The concatenation of two regular expressions is also a regular expression.
 - $R_1, R_2 \rightarrow R_1 \cdot R_2$

REGULAR EXPRESSION

Regular expressions are used for representing certain sets of strings in an algebraic fashion.

4. The iteration (closure) of a regular expression is also a regular expression.
 - $R \rightarrow R^*$
5. The regular expression over Σ are precisely those obtained recursively by the application of the above rules once or several times.

REGULAR EXPRESSION

Describe the following sets as regular expressions.

1. $\{0, 1, 2\}$
 $R = 0 + 1 + 2$
2. $\{abb, a, b, bba\}$
 $R = abb + a + b + bba$
3. $\{\epsilon, 0, 00, 000, \dots\}$
 $R = 0^*$
4. $\{1, 11, 111, 1111, \dots\}$
 $R = 1^+$

REGULAR EXPRESSION

Examples:

1. $R = (0 + 1)^*00(0 + 1)^*$
 $\{00, 10010, 010011, \dots\}$

2. $R = (1 + 10)^*$
 $\{\epsilon, 1, 110, \dots\}$

3. $R = (aa + ab + ba + bb)^+a$
 $\{aaa, aba, abbaa, \dots\}$

4. $R = ab^*a$
 $\{aa, aba, abba, abbba, \dots\}$

DESIGNING REGULAR EXPRESSION

Design regular expression for the following languages over $\{a, b\}$.

1. Language accepting strings of length exactly 2

$$L = \{aa, ab, ba, bb\}$$

$$\begin{aligned} R &= aa + ab + ba + bb \\ &= a(a + b) + b(a + b) \\ &= (a + b)(a + b) \end{aligned}$$

DESIGNING REGULAR EXPRESSION

Design regular expression for the following languages over $\{a, b\}$.

2. Language accepting strings of length at least 2

$$L = \{aa, ab, ba, bb, aaa, \dots\}$$

$$R = (a + b)(a + b)(a + b)^*$$

DESIGNING REGULAR EXPRESSION

Design regular expression for the following languages over $\{a, b\}$.

3. Language accepting strings of length at most 2

$$L = \{\epsilon, a, b, aa, ab, ba, bb\}$$

$$\begin{aligned} R &= \epsilon + a + b + aa + ab + ba + bb \\ &= (\epsilon + a + b)(\epsilon + a + b) \end{aligned}$$

REGULAR EXPRESSION

Describe the following languages as regular expressions.

1. $\{a, ab, abb, abbb, abbbb, \dots\}$
 $R = ab^*$

2. $\{\epsilon, ab, abab, ababab, \dots\}$
 $R = (ab)^*$

3. $\{a, c, ab, cb, abb, cbb, abbb, cbbb, abbbb, cbbbb, \dots\}$
 $R = (a + c)b^*$

4. $\{aaa, aab, aba, abb, baa, bab, bba, bbb\}$
 $R = (a + b)(a + b)(a + b)$

IDENTITIES OF REGULAR EXPRESSION

1. $\emptyset + R = R$
2. $\emptyset R = R\emptyset = \emptyset$
3. $\epsilon R = R\epsilon = R$
4. $\epsilon^* = \epsilon$ and $\emptyset^* = \epsilon$
5. $R + R = R$
6. $R^*R^* = R^*$
7. $RR^* = R^*R$
8. $(R^*)^* = R^*$
9. $\epsilon + RR^* = \epsilon + R^*R = R^*$
10. $(PQ)^*P = P(QP)^*$
11. $(P + Q)^* = (P^*Q^*)^* = (P^* + Q^*)^*$
12. $(P + Q)R = PR + QR$ and
 $R(P + Q) = RP + RQ$

EXAMPLE

Prove that

$$(1 + 00^*1) + (1 + 00^*1)(0 + 10^*1)^*(0 + 10^*1) = 0^*1(0 + 10^*1)^*$$

$$(1 + 00^*1) + (1 + 00^*1)(0 + 10^*1)^*(0 + 10^*1)$$

$$= (1 + 00^*1)[\epsilon + (0 + 10^*1)^*(0 + 10^*1)]$$

using I_{12}

$$= (1 + 00^*1)(0 + 10^*1)^*$$

using I_9

$$= (\epsilon 1 + 00^*1)(0 + 10^*1)^*$$

using I_3

$$= (\epsilon + 00^*)1(0 + 10^*1)^*$$

using I_{12}

$$= 0^*1(0 + 10^*1)^*$$

using I_9

ARDEN'S THEOREM

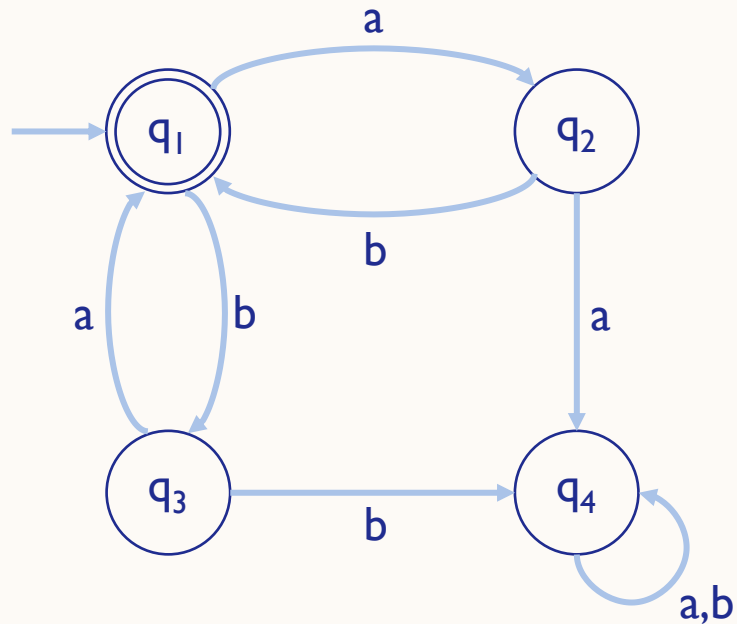
Let P and Q be two regular expressions over Σ . If P does not contain ϵ , then the following equation in R ,

$$R = Q + RP$$

has a unique solution given by $R = QP^*$

CONVERSION OF FSA TO REGULAR EXPRESSION

Find the regular expression recognized by the following FSA.



$$1) q_1 = \epsilon + q_2b + q_3a$$

$$2) q_2 = q_1a$$

$$3) q_3 = q_1b$$

$$4) q_4 = q_2a + q_3b + q_4a + q_4b$$

1)

$$q_1 = \epsilon + q_2b + q_3a \leftarrow 2) \text{ and } 3)$$

$$q_1 = \epsilon + q_1ab + q_1ba$$

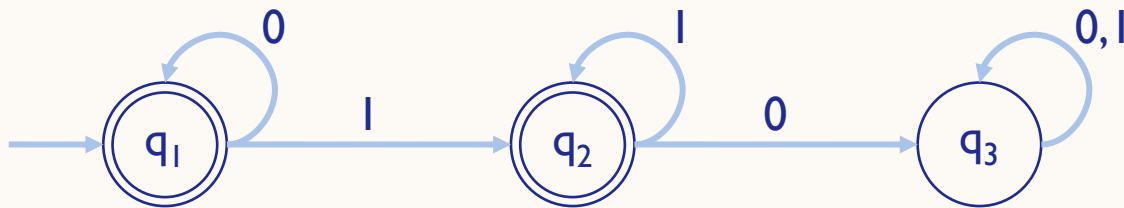
$$\frac{q_1}{\mathbf{R}} = \epsilon + \frac{q_1}{\mathbf{Q}} \frac{(ab + ba)}{\mathbf{P}} \leftarrow \text{Arden's Theorem}$$

$$q_1 = \epsilon(ab + ba)^* \leftarrow I_3$$

$$q_1 = (ab + ba)^*$$

CONVERSION OF FSA TO REGULAR EXPRESSION

Find the regular expression recognized by the following FSA.



$$1) q_1 = \epsilon + q_1 0$$

$$2) q_2 = q_1 1 + q_2 1$$

$$3) q_3 = q_2 0 + q_3 0 + q_3 1$$

Final State q_1

$$1) \quad \frac{q_1}{\overline{R}} = \frac{\epsilon}{\overline{Q}} + \frac{q_1 0}{\overline{R} \overline{P}} \leftarrow \text{Arden's Theorem}$$

$$q_1 = \epsilon 0^* \leftarrow I_3$$

$$q_1 = 0^* \rightarrow 4)$$

Final State q_2

$$2) \quad q_2 = q_1 1 + q_2 1 \leftarrow 4)$$

$$\frac{q_2}{\overline{R}} = \frac{0^* 1}{\overline{Q}} + \frac{q_2 1}{\overline{R} \overline{P}} \leftarrow \text{Arden's Theorem}$$

$$q_2 = 0^* 1 1^*$$

Union of Final States q_1 and q_2

$$R = q_1 + q_2$$

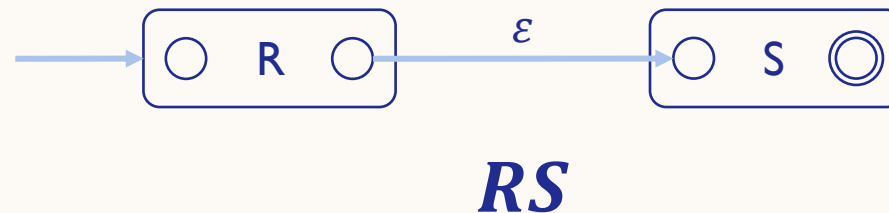
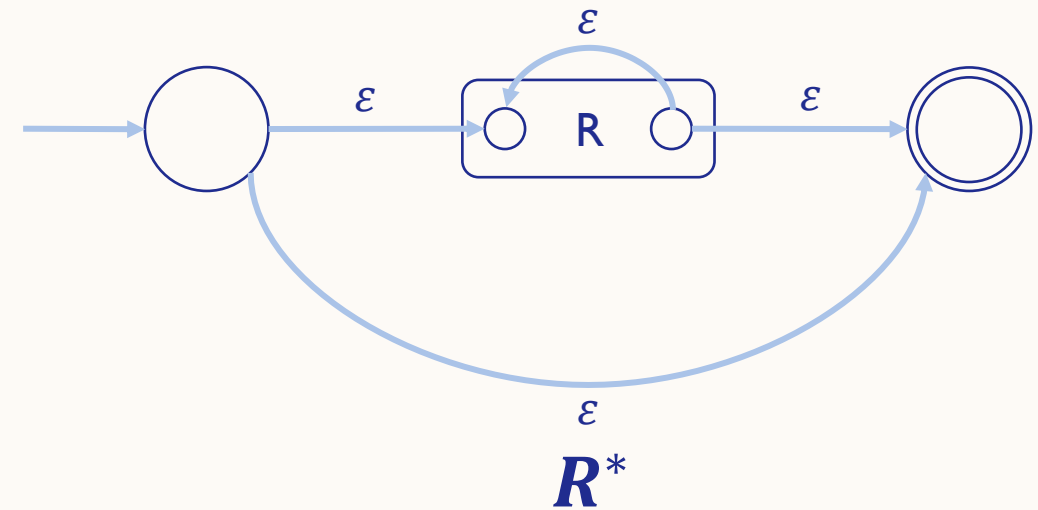
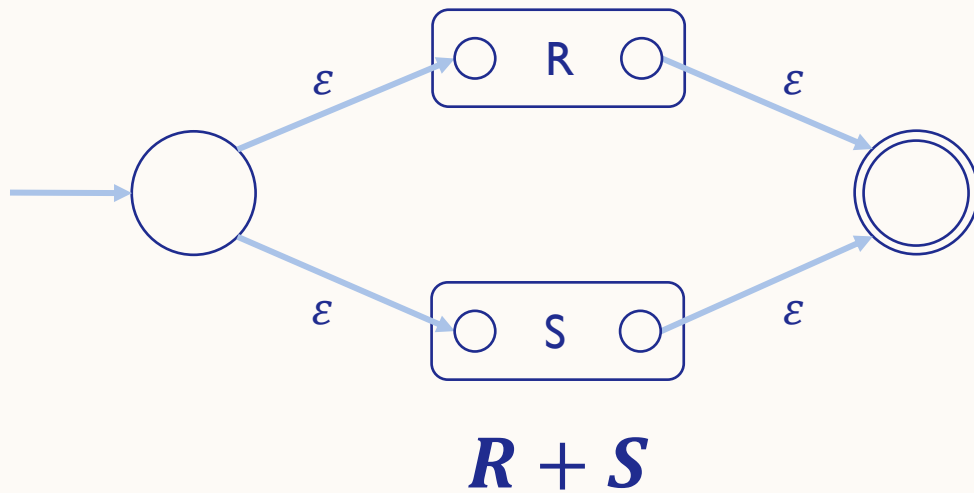
$$R = 0^* + 0^* 1 1^*$$

$$R = 0^* (\epsilon + 1 1^*) \leftarrow I_9$$

$$R = 0^* 1^*$$

CONVERSION OF REGULAR EXPRESSION TO FSA

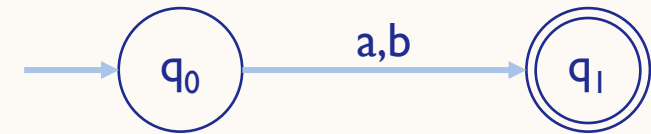
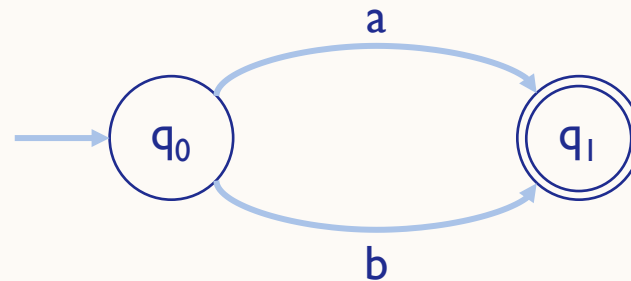
There is an NFA with ϵ -transitions (ϵ -NFA) accepting the same language defined by a regular expression.



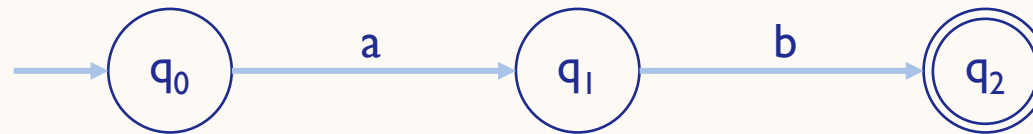
CONVERSION OF REGULAR EXPRESSION TO FSA

If the regular expression is simple enough, the NFA can be constructed directly without going through the ϵ -NFA.

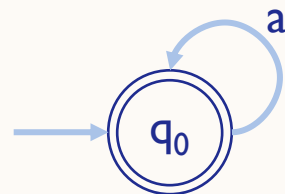
$a + b$



ab



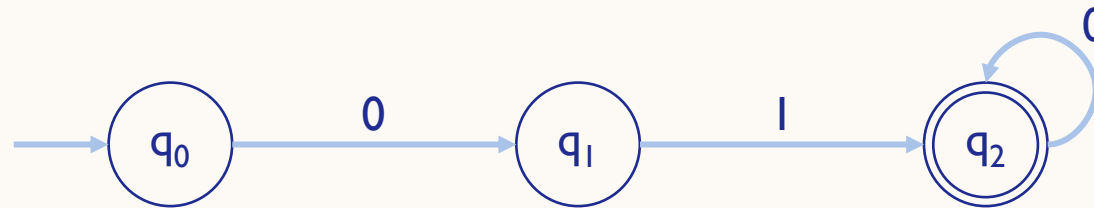
a^*



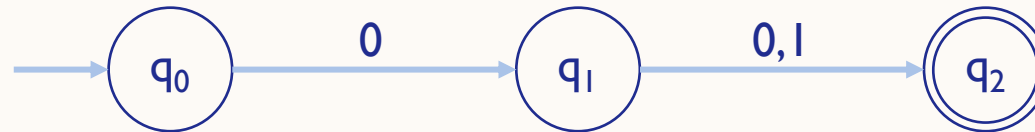
EXAMPLES

Convert the following regular expressions to their equivalent Finite Automata, or vice versa.

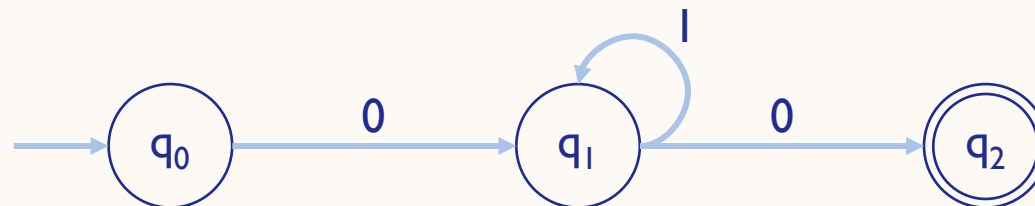
010^*



$0(0 + 1)$

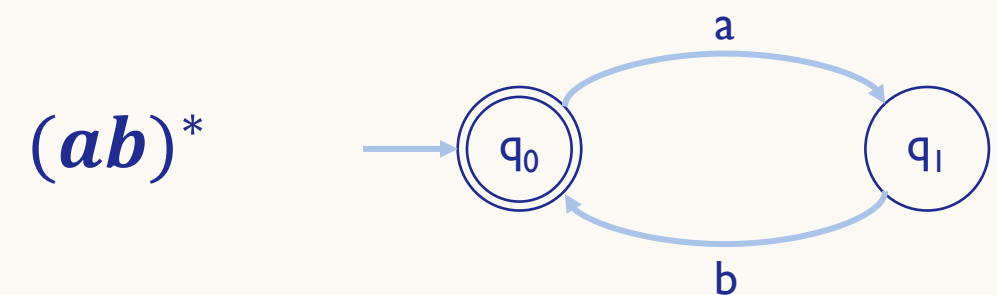
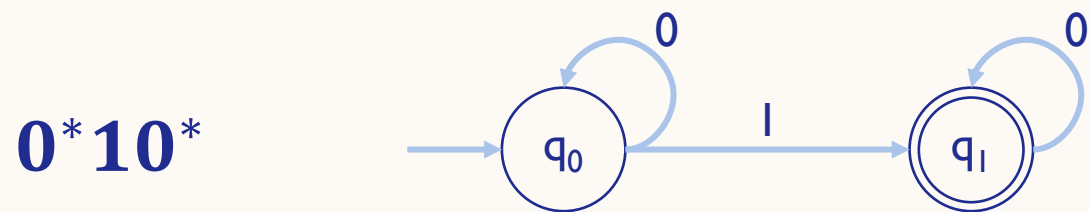
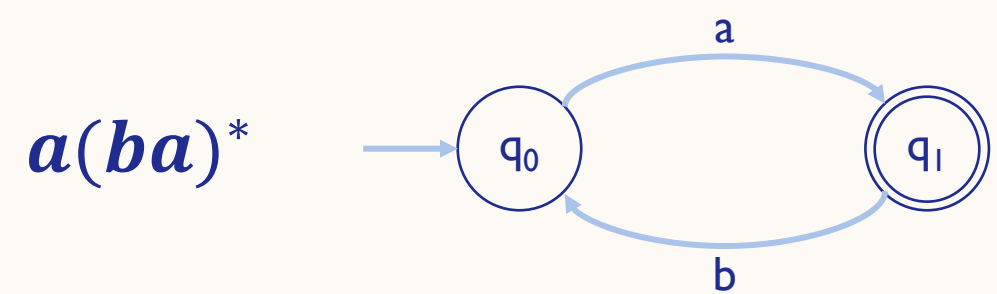
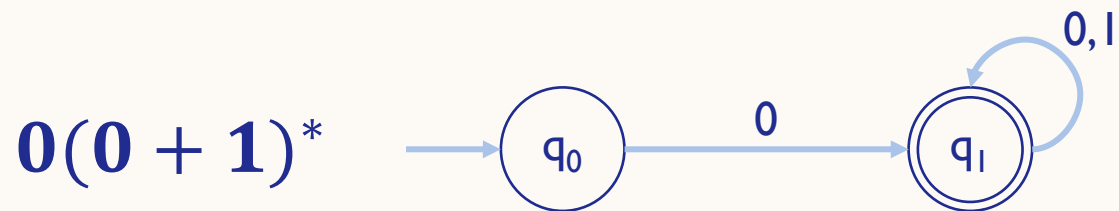


01^*0

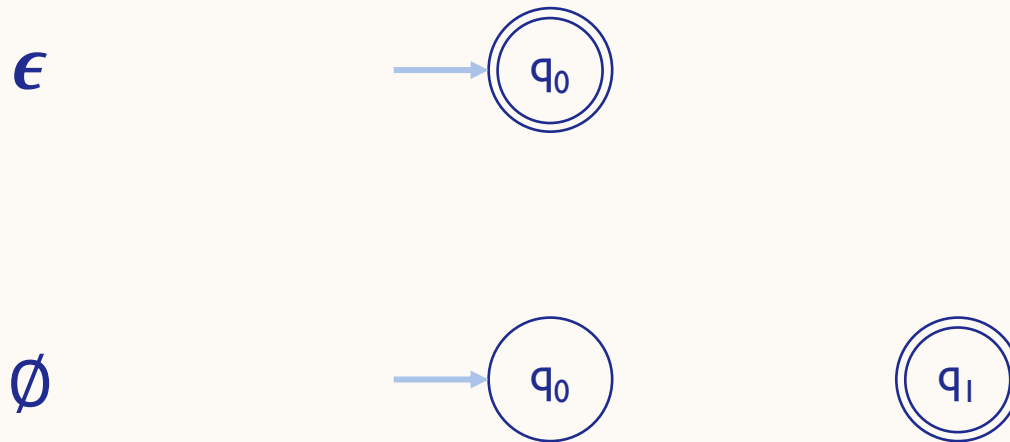


EXAMPLES

Convert the following regular expressions to their equivalent Finite Automata, or vice versa.



CONVERSION OF REGULAR EXPRESSION TO FSA





PRACTICE

EXERCISES

- I. Write the regular expressions for the following languages.
 - a. The set of strings over alphabet $\{0,1\}$ ending in 010
 - b. The set of strings over alphabet $\{a,b\}$ beginning with aab
 - c. The set of strings over alphabet $\{0,1\}$ beginning with 1 and ending with 0
 - d. $\{00, 0000, 000000, 00000000, \dots\}$
 - e. The set of strings over alphabet $\{0,1\}$ containing 101
 - f. The set of strings over alphabet $\{0,1\}$ in which the third last symbol is 1
 - g. The set of strings over alphabet $\{a,b,c\}$ containing at least one a and one b

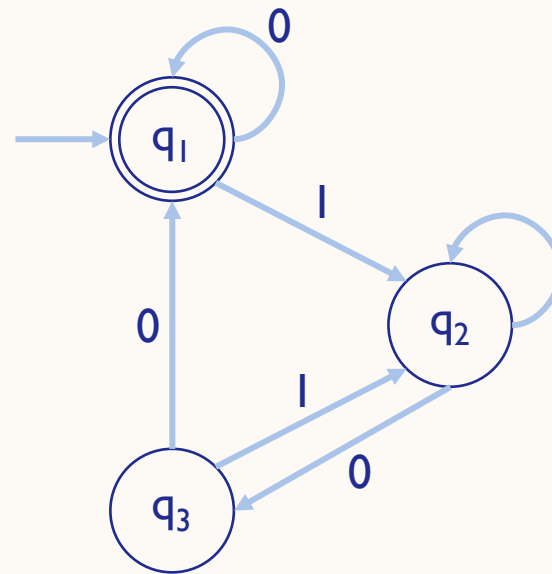
EXERCISES

2. Write the regular expression for representing the set of strings over alphabet $\{0,1\}$ in which every 0 is immediately followed by at least two 1s.

Prove that the regular expression $R = \epsilon + 1^*(011)^*(1^*(011)^*)^*$ also describes the same set of strings.

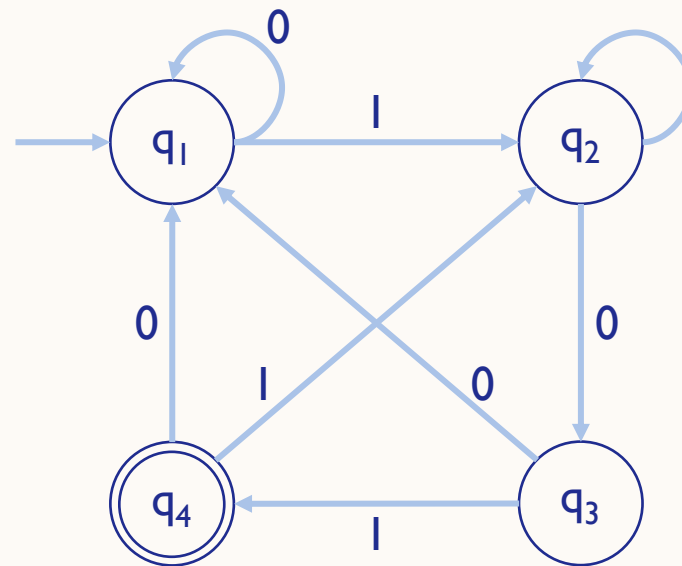
EXERCISES

3. Construct the regular expression corresponding to the following FSA.



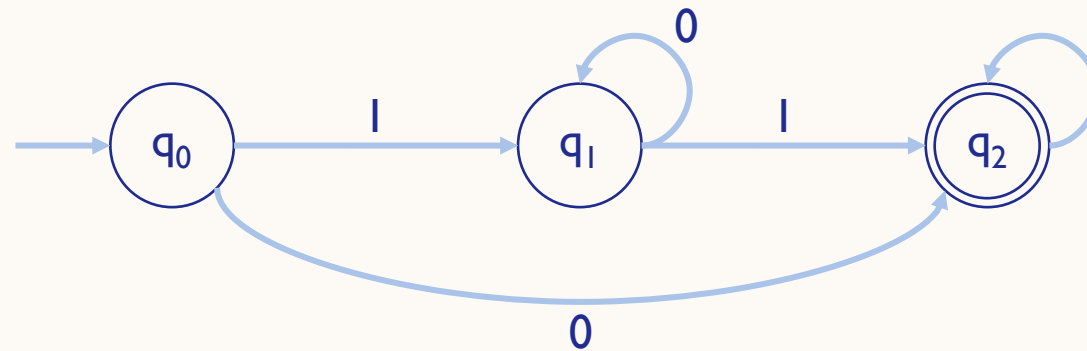
EXERCISES

4. Construct the regular expression corresponding to the following FSA.



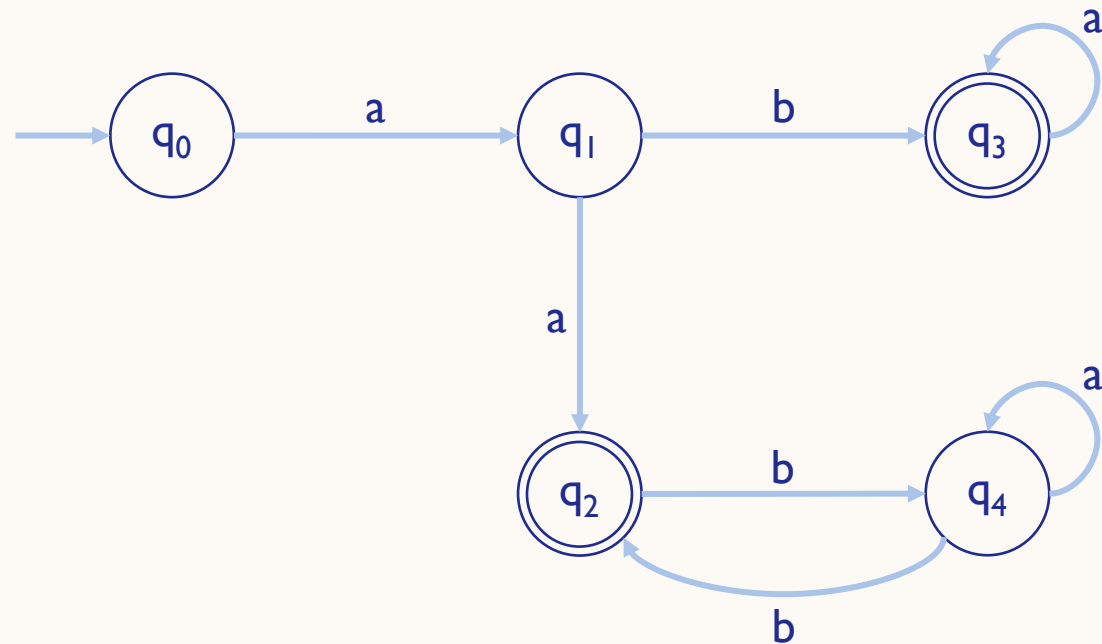
EXERCISES

5. Find the regular expression corresponding to the following FSA.



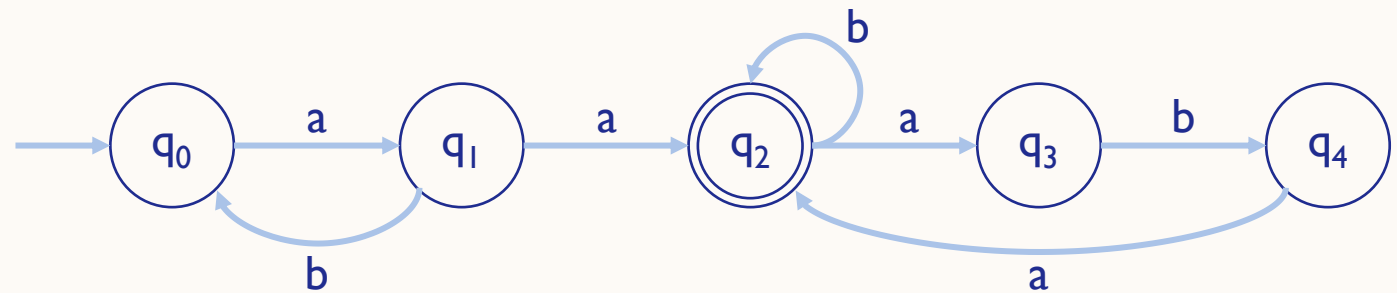
EXERCISES

6. Find the regular expression corresponding to the following FSA.



EXERCISES

7. Find the regular expression corresponding to the following FSA.



EXERCISES

8. Convert the following regular expressions to their equivalent FSA.

a. $a(bc)^*$

b. $(a + b)^*(abb + a^+b)$

c. $(0 + 1)^*(00 + 11)(0 + 1)^*$

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

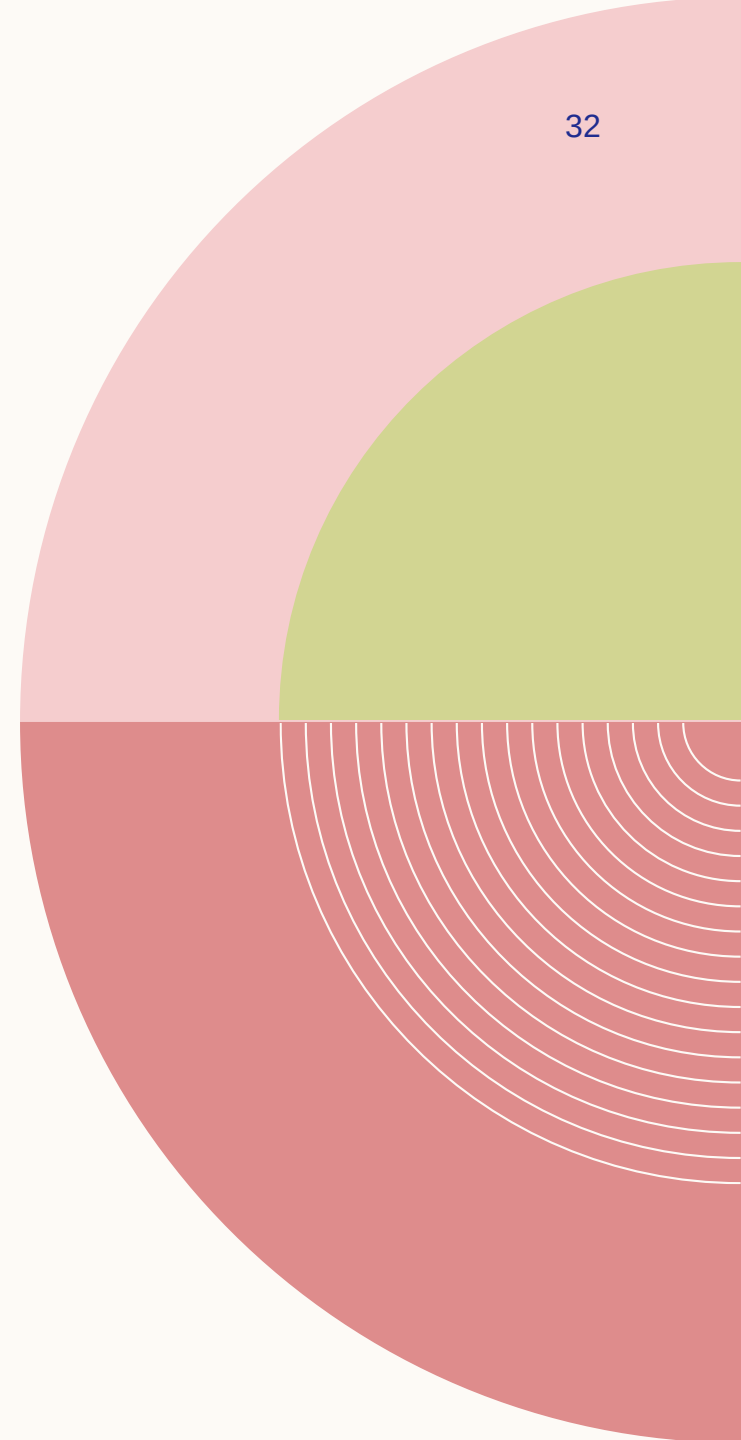
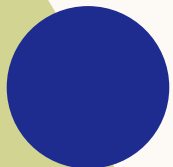
NEXT

Production Rules of FSA:

Regular Grammar

Conversion of FSA to Regular Grammar

Conversion of Regular Grammar to FSA



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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